RoboSoc Workshop 1 Terminology

Voltage (potential difference)
The electrical push between two points, measured in volts (V).

Current
The flow of electric charge, measured in amperes (A).

Resistance
Opposition to current flow, measured in ohms (Ω).

Resistor
Limits current to protect the LED. We use 330 Ω resistors.

LED
A light-emitting diode that lights when current flows the right way.

Anode (+)
The LED’s positive side, usually the longer leg.

Cathode (−)
The LED’s negative side, usually the shorter leg and flat side.

Power rail
A line of connected breadboard holes that distributes power.

Series
Components along one path, carrying the same current.

Parallel
Separate branches across the same two points, sharing the same voltage.

AND logic
The LED lights only when both buttons are pressed.

OR logic
The LED lights when one or both buttons are pressed.